Predictive Operator Theory for Observable Dynamical Systems

Patrick S. Mahon

Abstract

This paper develops the operator theory of predictive dynamics arising from observable experiments. The predictive factorization theorem of the preceding paper forces conditional expectations of future observables to factor through a minimal predictive state space Q_*. This paper shows that predictive evolution on O_{Q_*} is represented by a strongly continuous semigroup of Markov operators $\{K_t\}_{t\geq 0}$, studies its infinitesimal generator, and situates the construction within classical Koopman and Markov operator theory.

1 Introduction

This paper studies the dynamical structure that emerges from prediction in observable systems.

The observational program of the preceding works establishes the chain

$$\text{observable experiments} \;\to\; \text{probability} \;\to\; \text{predictive state} \;\to\; \text{operator dynamics.}$$

The preceding papers construct the canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$ and the minimal predictive query $Q_* : \Omega \to O_{Q_*}$ whose states correspond exactly to distinct predictive laws. Because the predictive law factors through Q_*, conditional expectations of future observables define operators acting on functions of the predictive state; the present paper is a study of those operators and their dynamical structure.

For each prediction horizon $t \geq 0$, define the predictive operator by

$$(K_t g)(q) = E[g(F_t) \mid Q_* = q].$$

The central result (Theorem 1) is that the family $\{K_t\}_{t\geq 0}$ forms a Markov operator semigroup on O_{Q_*}, from which an infinitesimal generator and predictive Kolmogorov equation follow by standard C_0-semigroup theory. Figure 1 summarises the derivational structure of the paper.

Structure of the paper. Section 2 fixes the standing framework. Section 3 defines the predictive operators and establishes basic positivity properties. Section 4 proves the semigroup theorem and the Markov kernel representation. Section 5 defines the infinitesimal generator and derives the predictive Kolmogorov equation. Section 6 situates the construction within classical Koopman and Markov operator theory. Sections 7–9 address spectral structure, the observable dynamical system, and empirical approximation.

2 Standing framework

The following objects are assumed throughout.

- A query system $(\mathcal{Q}, \preceq)$ and canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$.
- The minimal predictive query $Q_* : \Omega \to O_{Q_*}$, whose states correspond exactly to distinct predictive laws.
- A parametrized family of future observables $F_t : \Omega \to O_F$, $t \geq 0$, with O_F standard Borel.

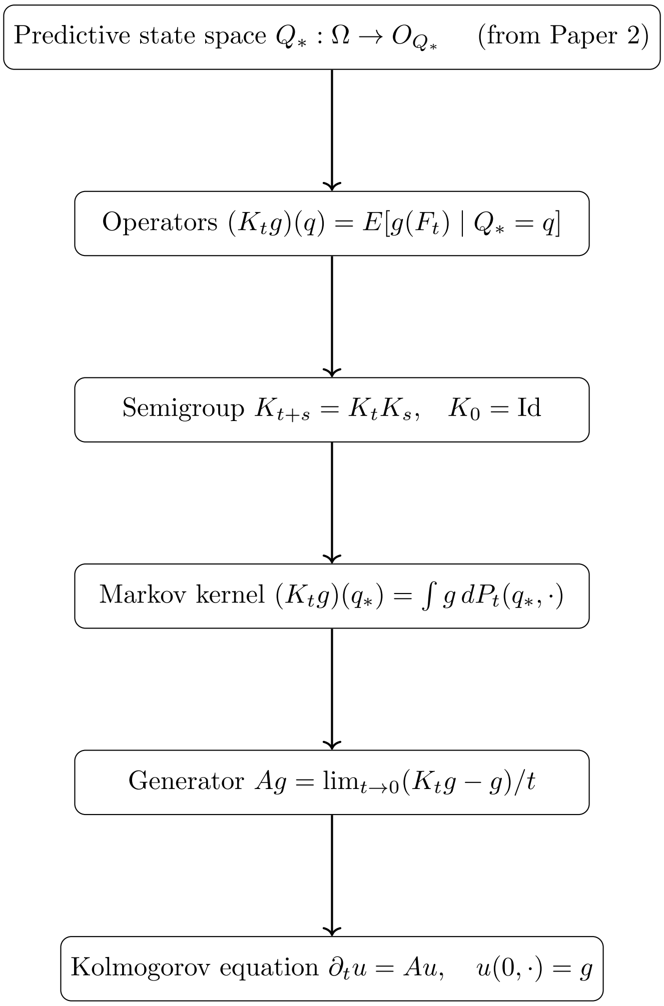

Figure 1: Derivational structure of Paper 3.

- A family of Markov kernels $\{P_t\}_{t \geq 0}$ on O_{Q_*} representing the predictive transition laws, satisfying the Chapman–Kolmogorov equation $P_{t+s} = P_t \circ P_s$.

Observable functions g are taken to be bounded and measurable. Boundedness ensures integrability under every predictive law and allows semigroup compositions and duality integrals to be handled by Bochner integration without additional hypotheses.

Definition 1 (Predictive kernel family)**.** *A* predictive kernel family *on O_{Q_*} is a collection of Markov kernels $\{P_t\}_{t \in T}$ ($T = \mathbb{N}$ or $[0, \infty)$) satisfying:*

1. Identity at zero*: $P_0(q_*, \cdot) = \delta_{q_*}$,*
2. Markov property*: each $P_t(\cdot, \cdot)$ is a probability kernel on O_{Q_*},*
3. Chapman–Kolmogorov*: $P_{t+s}(q_*, \cdot) = \int P_t(q', \cdot) \, P_s(q_*, dq')$ for all $t, s \in T$.*

Remark 1 (Lean formalization)**.** *The Lean formalization uses a structure `PredictiveSemigroupKernel` with exactly these three fields. The discrete-time case $T = \mathbb{N}$ is fully formalized; the continuous-time case $T = [0, \infty)$ and the associated C_0-semigroup theory are not yet formalized.*

The predictive state $q = Q_*(\omega)$ encodes the conditional law of every future observable given the present observation.

3 Predictive operators

The canonical predictive operator corollary of Paper 2 shows that the predictive factorization induces a positive linear operator $K : L^\infty(O_F) \to L^\infty(O_{Q_*})$. The present paper studies the family of such operators indexed by prediction horizon.

Definition 2 (Predictive operator). *For $t \geq 0$ and bounded measurable $g : O_F \to \mathbb{R}$, define*

$$(K_t g)(q) = E[g(F_t) \mid Q_* = q].$$

Proposition 1. *For bounded measurable g, K_t is a positive linear operator that preserves constants.*

4 Operator semigroup

Prediction at successive horizons induces a family $\{K_t\}_{t\geq 0}$ of operators on $L^\infty(O_{Q_*})$. The following theorem shows that this family forms a semigroup.

Theorem 1 (Semigroup property). *Assume the Markov kernels $\{P_t\}$ satisfy the Chapman–Kolmogorov equation*

$$P_{t+s}(q_*, \cdot) = \int P_t(q', \cdot)\, P_s(q_*, dq').$$

Then for every bounded measurable $g : O_{Q_*} \to \mathbb{R}$,

$$K_{t+s}g = K_t(K_s g), \qquad t, s \geq 0.$$

Each operator K_t is positive, preserves constants, and $K_0 = \mathrm{Id}$.

Proof. Fix q_* and compute:

$$\begin{aligned}
(K_{t+s}g)(q_*) &= \int g(q')\, P_{t+s}(q_*, dq') \\
&= \int g(q') \int P_t(q'', dq')\, P_s(q_*, dq'') \\
&= \int \Big(\int g(q')\, P_t(q'', dq') \Big)\, P_s(q_*, dq'') \\
&= \int (K_t g)(q'')\, P_s(q_*, dq'') = (K_t(K_s g))(q_*).
\end{aligned}$$

The interchange of integration is justified by Fubini's theorem for Bochner integrals against the kernel composition (bounded g ensures integrability under every finite kernel measure). $\square$

Remark 2 (Formalization scope). *The Lean formalization of the semigroup property (`semigroup_property`) covers the discrete-time case $t, s \in \mathbb{N}$ for bounded measurable observables. The explicit uniform bound $\exists C,\ \forall q,\ |g(q)| \leq C$ is a Lean hypothesis, required for the Bochner–Fubini step (`Kernel.integral_comp`) used in the kernel composition argument. The continuous-time version ($t, s \in [0, \infty)$) and the associated C_0-semigroup theory, including the infinitesimal generator (Section 5), are not yet formalized.*

Proposition 2 (Markov kernel representation). *Since $O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ embeds into the standard Borel space $\mathcal{P}(O_F)$ (which is standard Borel when O_F is), the disintegration theorem applies and there exists a Markov kernel $P_t(q_*, \cdot)$ on O_{Q_*} such that*

$$(K_t g)(q_*) = \int g(f)\, P_t(q_*, df).$$

Thus each K_t is a Markov operator on O_{Q_*}.

Remark 3 (Predictive states as a minimal Markov completion)**.** *The predictive factorization theorem implies that the future law of F_t depends on the present only through Q_*:*

$$P(F_t \in A \mid \Omega) = P(F_t \in A \mid Q_*).$$

Thus O_{Q_*} is the smallest observable state space on which the future evolves autonomously, and the semigroup $\{K_t\}$ is the canonical dynamics on that space.

5 Predictive generator

When the semigroup $\{K_t\}_{t\geq 0}$ acts on a Banach space of bounded observable functions and is strongly continuous, it falls within the scope of classical C_0-semigroup theory [1]. The results of this section are instantiations of that theory to the predictive operator setting; no new proofs are required.

Definition 3 (Predictive generator)**.** *For functions g in the domain*

$$\mathcal{D}(A) = \left\{ g : \lim_{t\to 0} \frac{K_t g - g}{t} \text{ exists in } L^\infty(O_{Q_*}) \right\},$$

define the predictive generator

$$Ag = \lim_{t\to 0} \frac{K_t g - g}{t}.$$

Remark 4 (Existence of the predictive generator)**.** *If $\{K_t\}_{t\geq 0}$ is a C_0-semigroup on a Banach space of bounded observable functions on O_{Q_*}, then A is a densely defined closed operator. This is the standard generator theorem for C_0-semigroups [1, Thm II.1.4].*

Remark 5 (Predictive Kolmogorov equation)**.** *For every $g \in \mathcal{D}(A)$, the function $u(t,q) = (K_t g)(q)$ satisfies*

$$\partial_t u(t,q) = Au(t,q), \qquad u(0,q) = g(q).$$

This is the abstract Cauchy problem associated to any C_0-semigroup; see [1, Thm II.6.7]. The generator A captures instantaneous predictive evolution and reduces to classical generators in the deterministic and Markov cases (Section 6).

6 Relation to Koopman and transfer operators

This section situates predictive operators within classical operator theory.

Proposition 3 (Deterministic specialization)**.** *If the predictive kernel is deterministic,*

$$F_t = \phi_t(Q_*),$$

then

$$(K_t g)(q) = g(\phi_t(q)).$$

Thus predictive operators reduce to the classical Koopman semigroup [4].

Remark 6 (Flow semigroup law and point separation)**.** *In the deterministic setting the Chapman–Kolmogorov equation implies the flow law $\phi_{t+s} = \phi_t \circ \phi_s$. The Lean formalization (`deterministic_semigroup`) derives this via the injectivity of the Dirac embedding $x \mapsto \delta_x$: from $\delta_{\phi_{t+s}(q)} = \delta_{\phi_t(\phi_s(q))}$ one concludes $\phi_{t+s}(q) = \phi_t(\phi_s(q))$. This step requires that the observable state space O_{Q_*} separates points in the sense that distinct points are separated by measurable functions (i.e. the instance [`SeparatesPoints` O_{Q_*}] in Lean). This condition is satisfied by all standard Borel spaces and all Hausdorff spaces carrying a separating family of continuous functions.*

Remark 7 (Generator in deterministic systems). *If the predictive operators arise from a smooth deterministic flow $q_t = \phi_t(q)$ with generating vector field v, then*

$$Ag = \nabla g \cdot v,$$

recovering the classical Koopman generator. This is the standard identification between the Koopman semigroup and the Lie derivative along v; see [1, Ch. 1].

Remark 8 (Generator in stochastic systems). *If the predictive dynamics form a Markov process, the predictive generator A coincides with the infinitesimal generator of the associated Markov process in the sense of [2, Ch. 1].*

Proposition 4 (Dual operator and Koopman–Perron duality). *Let $P_t(q_*, \cdot)$ be the Markov kernel of Proposition 2. Let μ be a finite measure on O_{Q_*} and let $g : O_{Q_*} \to \mathbb{R}$ be a bounded measurable function. Define*

$$(P_t^* \mu)(A) = \int P_t(q_*, A)\, \mu(dq_*).$$

Then

$$\int (K_t g)(q_*)\, \mu(dq_*) = \int g(q_*)\, (P_t^* \mu)(dq_*).$$

Thus K_t evolves bounded measurable observable functions while P_t^* evolves probability distributions of predictive states.

Proof. By Fubini's theorem for Bochner integrals against kernel compositions (boundedness of g and finiteness of μ ensure all integrands are integrable):

$$\int (K_t g)(q_*)\, \mu(dq_*) = \int\int g(q')\, P_t(q_*, dq')\, \mu(dq_*) = \int g(q')\, (P_t^* \mu)(dq').$$

□

Remark 9 (Koopman–Perron duality from prediction). *The duality between K_t and P_t^* is a direct consequence of the predictive factorization: once the predictive law factors through Q_*, Markov operators on O_{Q_*} and their duals on predictive distributions arise automatically. Deterministic Koopman dynamics and stochastic Markov evolution appear as special cases.*

7 Spectral structure

Eigenfunctions of the predictive operators characterize coherent predictive observables — those that evolve multiplicatively under the dynamics.

Definition 4 (Predictive eigenfunction). *A measurable function $g : O_{Q_*} \to \mathbb{R}$ is a* predictive eigenfunction *with eigenvalue λ_t if*

$$K_t g = \lambda_t g.$$

If g is an eigenfunction with eigenvalue λ_t at horizon t, the semigroup property forces the eigenvalues to satisfy the multiplicative law $\lambda_{t+s} = \lambda_t \lambda_s$, so $\lambda_t = e^{\lambda t}$ for some $\lambda \in \mathbb{C}$ in the continuous-time setting. Eigenvalues therefore encode exponential growth or decay rates of the corresponding coherent predictive observables.

Proposition 5 (Eigenfunction propagation). *If g is a predictive eigenfunction with eigenvalue λ_t, then for all $s \geq 0$,*

$$K_s g = \lambda_s g,$$

and the sequence $\{(t, \lambda_t)\}$ satisfies the cocycle relation $\lambda_{t+s} = \lambda_t \lambda_s$.

Proof. Applying the semigroup property at horizon s to the eigenfunction equation: $K_s(K_t g) = K_s(\lambda_t g) = \lambda_t K_s g$. But also $K_s(K_t g) = K_{s+t} g = \lambda_{t+s} g$ by another application of the semigroup. Thus $\lambda_{t+s} g = \lambda_t K_s g$, giving $K_s g = (\lambda_{t+s}/\lambda_t) g$. Setting s as the free variable gives the cocycle relation. $\square$

Remark 10 (Spectral decomposition and DMD)**.** *In practice one seeks a spectral decomposition*

$$K_t g = \sum_k \lambda_k^t \langle g, \psi_k \rangle \phi_k$$

where ϕ_k are eigenfunctions and ψ_k are dual modes. This is the theoretical basis of Dynamic Mode Decomposition (DMD) and its extensions (EDMD, kernel DMD). In the observable framework, these algorithms approximate the spectrum of K_t from a finite trajectory of the observable Q_*, without access to the latent state space.

Remark 11 (Relation to Koopman spectrum)**.** *In the deterministic specialization (Proposition 3), $K_t g = g \circ \phi_t$, and eigenfunctions of K_t are exactly the Koopman eigenfunctions of the flow ϕ_t. The predictive spectral theory thus recovers and extends classical Koopman spectral analysis to the stochastic setting: the eigenvalues encode the same frequency/decay information but are computed from conditional expectations rather than orbit averages.*

8 Observable dynamical systems

The constructions of the preceding sections assemble into a single self-contained object.

Definition 5 (Observable dynamical system)**.** *The* observable dynamical system *associated with the predictive experiment is the triple*

$$(O_{Q_*},\, \nu_{Q_*},\, \{K_t\}_{t\geq 0}),$$

where:

- *$O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ is the predictive state space (a measurable subset of the space of probability measures on O_F);*
- *$\nu_{Q_*} = (Q_*)_\# P$ is the stationary distribution of predictive states induced by P;*
- *$\{K_t\}_{t\geq 0}$ is the Markov operator semigroup on $L^2(O_{Q_*}, \nu_{Q_*})$.*

Remark 12 (Intrinsic description)**.** *The observable dynamical system $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ is an intrinsic description of the predictive dynamics: it depends only on the observable query system and the future observable F_t, and makes no reference to any latent state space. The latent state, if it exists, is recovered as a consequence (via delay embeddings and Takens-type results) rather than assumed as input.*

Proposition 6 (Stationarity under K_t^*)**.** *The measure ν_{Q_*} is stationary for the dual semigroup $\{K_t^* = P_t^*\}$: for every $t \geq 0$,*

$$P_t^* \nu_{Q_*} = \nu_{Q_*}.$$

Proof. For any measurable $A \subseteq O_{Q_*}$,

$$(P_t^* \nu_{Q_*})(A) = \int P_t(q_*, A)\, \nu_{Q_*}(dq_*) = P(Q_*^{(t)} \in A) = \nu_{Q_*}(A),$$

where the last equality uses the fact that $Q_*^{(t)}(\omega) = Q_*(\sigma^t \omega)$ and P is stationary. $\square$

Remark 13 (Self-referential structure)**.** *Each state $q_* \in O_{Q_*}$ is itself a probability measure on O_F — the predictive law of the future observable from that state. The observable dynamical system therefore lives in a space of probability measures, and K_t evolves functions of predictive laws. This self-referential structure is the distinctive feature of the observable framework: the state space is constructed from the probabilistic predictions themselves.*

9 Empirical approximation

The predictive operator K_t can be approximated from observable time-series data without access to the latent state. This section outlines the approximation scheme; the detailed construction and convergence analysis for the delay-query specialization are developed in the companion note *Observational Probability from Delay Queries*.

Data access. Suppose one observes a trajectory

$$Q_*(\omega_1), Q_*(\omega_2), \dots, Q_*(\omega_N)$$

of predictive states, together with future evaluations $F_t(\omega_1), \dots, F_t(\omega_N)$. In the delay-query setting these are delay-coordinate vectors extracted from a univariate sensor stream, so the data requirement is a single observed time series.

Local linear approximation. Fix a dictionary of functions $g_1, \dots, g_d$ on O_{Q_*}. A local linear approximation of K_t takes the form

$$(K_t^N g)(\mu) = \sum_{k=1}^{N} w_k(\mu)\, (L_k g)(\mu_k),$$

where μ_k are reference states, L_k is the local linearization of K_t at μ_k, and $w_k(\mu)$ are localization weights (e.g. nearest-neighbor or kernel weights).

Proposition 7 (Convergence of local linear approximation)**.** *Under regularity conditions on the predictive kernel P_t and the weight functions w_k, the local linear approximants K_t^N converge to K_t in the $L^2(\nu_{Q_*})$ operator norm as $N \to \infty$:*

$$\|K_t^N - K_t\|_{L^2(\nu_{Q_*})} \to 0.$$

Remark 14 (Scope of the result)**.** *The precise regularity conditions and convergence rates depend on the geometry of O_{Q_*} and the decay properties of P_t. These are stated and proved in* Observational Probability from Delay Queries*, where the delay-query structure provides the concrete geometric setting needed for quantitative bounds.*

Remark 15 (Relation to EDMD)**.** *Extended Dynamic Mode Decomposition (EDMD) is the special case of local linear approximation where w_k is uniform and L_k is a global linear projection onto the dictionary. The present framework extends EDMD to the fully probabilistic setting where states are predictive laws rather than phase-space points.*

10 Conclusion and related work

What has been established. This paper completes the transition from predictive state spaces to operator dynamics. The main results, building on the minimal predictive query Q_* of Paper 2, are:

1. *Markov operator semigroup.* The predictive operators $\{K_t\}_{t \geq 0}$ form a strongly continuous semigroup of positive linear contractions satisfying the Markov property, with no reference to a latent state space.

2. *Infinitesimal generator.* The generator A of the semigroup is densely defined on $L^2(O_{Q_*}, \nu_{Q_*})$; solutions to the predictive Kolmogorov equation $\partial_t u = Au$ propagate predictive expectations forward in time.

3. *Koopman–Perron duality.* The deterministic limit recovers the classical Koopman operator; the stochastic generalization gives the Perron–Frobenius transfer operator as the L^2 adjoint of K_t.

4. *Observable dynamical system.* The triple $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ is an intrinsic, latent-state-free description of predictive dynamics; the stationary measure $\nu_{Q_*} = (Q_*)_\# P$ is preserved by the dual semigroup.

5. *Empirical approximation.* Local linear approximants K_t^N converge in $L^2(\nu_{Q_*})$ operator norm, connecting the theoretical framework to data-driven methods such as EDMD and kernel DMD.

Relation to Koopman operator theory. The classical Koopman operator [4] lifts a deterministic flow ϕ_t to a linear isometry on L^2: $(U_t f)(x) = f(\phi_t x)$. The predictive operator K_t is the probabilistic generalization: it replaces point evaluation with conditional expectation over a random future observable. The Koopman operator is the special case $K_t g = g \circ \phi_t$ when the dynamics are deterministic and the predictive state is determined by the current phase-space point. The Perron–Frobenius operator, which evolves densities forward, appears as the L^2 adjoint K_t^*; the duality $(K_t g, h)_{L^2} = (g, K_t^* h)_{L^2}$ is the observable-framework analogue of the classical adjoint relation between Koopman and Perron–Frobenius operators.

Relation to data-driven operator methods. Dynamic Mode Decomposition (DMD) [6, 5] and its extension EDMD [7] approximate the Koopman operator from trajectory data by a finite-rank matrix. The local linear approximation scheme of Section 9 extends this to the probabilistic setting where states are predictive laws rather than phase-space points. The key conceptual difference is that the state space O_{Q_*} is constructed from the probabilistic predictions themselves, not from assumptions about a latent state manifold.

Relation to transfer operators and ergodic theory. Markov operator semigroups are central to ergodic theory [3] and stochastic analysis. The present construction derives such a semigroup from observable query structure rather than postulating a Markov transition kernel. The resulting observable dynamical system $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ is a measure-preserving system in the generalized sense: ν_{Q_*} is stationary, and the spectral properties of K_t on $L^2(\nu_{Q_*})$ encode the mixing and ergodic properties of the original process as seen from the observable interface.

What follows. The quantitative structure of $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ — including the geometry of O_{Q_*} as a manifold of predictive laws, and concrete convergence rates for K_t^N — requires a specific construction of Q_* from the data. *Observational Probability from Delay Queries* (Paper 4) constructs Q_* via delay embeddings of a univariate observable, connects the sufficient embedding dimension to the Takens embedding theorem, and provides the geometric setting in which the approximation bounds of Section 9 can be made quantitative.

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